

ACTG - Prior analysis

We have

$$\begin{aligned}
p(m | D) &= \frac{Z_m(D)p(m)}{\sum_{j \in \mathcal{M}} Z_j(D)p(j)}, \\
&= \frac{Z_m(D) \frac{Z_j(D_0)}{\sum_{k \in \mathcal{M}} Z_k(D_0)}}{\sum_{j \in \mathcal{M}} Z_j(D) \frac{Z_m(D_0)}{\sum_{k \in \mathcal{M}} Z_k(D_0)}}, \\
&= \frac{Z_m(D)Z_m(D_0)}{\sum_{j \in \mathcal{M}} Z_j(D)Z_j(D_0)}, \\
&= \frac{\int L(\beta^{(m)}|D)\pi(\beta^{(m)}, a_0 | D_0) d\beta^{(m)} da_0 \int L(\beta^{(m)}|D_0)\pi_0(\beta^{(m)} | \psi_0) d\beta^{(m)}}{\sum_{j \in \mathcal{M}} \int L(\beta^{(j)}|D)\pi(\beta^{(j)}, a_0 | D_0) d\beta^{(j)} da_0 \int L(\beta^{(j)}|D_0)\pi_0(\beta^{(j)} | \psi_0) d\beta^{(j)}}, \\
&= \frac{\int L(\beta^{(m)}|D) \frac{L(\beta^{(m)}|D_0)^{a_0}}{C(a_0)} \pi_A(a_0) \pi_0(\beta^{(m)} | \tau_0) d\beta^{(m)} da_0 \int L(\beta^{(m)}|D_0)\pi_0(\beta^{(m)} | \psi_0) d\beta^{(m)}}{\sum_{j \in \mathcal{M}} \int L(\beta^{(j)}|D) \frac{L(\beta^{(j)}|D_0)^{a_0}}{C(a_0)} \pi_A(a_0) \pi_0(\beta^{(j)} | \tau_0) d\beta^{(j)} da_0 \int L(\beta^{(j)}|D_0)\pi_0(\beta^{(j)} | \psi_0) d\beta^{(j)}},
\end{aligned}$$

where $a_0 \sim \text{Beta}(\delta_0, \lambda_0)$, $\beta^{(m)} | \psi_0 \sim N(\mathbf{0}, \phi_0^2 \cdot \mathbf{I}_{p^{(m)}})$ and $\beta^{(m)} | \tau_0 \sim N(\mathbf{0}, \tau_0^2 \cdot \mathbf{I}_{p^{(m)}})$.

Prior sensitivity

We start by computing probabilities for several values of ψ_0 , $(\delta_0, \lambda_0) = (1, 1)$ and $\tau_0 = \sqrt{5}$:

m	$p(m)$	$\log(p(D m))$	$p(m D)$	ψ_0
age, treatment, race, cd4	0.802	-35.854	0.815	0.250
age, treatment, race, cd4	0.554	-35.876	0.552	0.500
age, treatment, cd4	0.487	-35.869	0.488	1.000
age, treatment, cd4	0.295	-35.846	0.303	2.000

Then, we compute probabilities for several values of τ_0 , $(\delta_0, \lambda_0) = (1, 1)$ and $\psi_0 = \sqrt{5}$:

m	$p(m)$	$\log(Z_m(D))$	$p(m D)$	τ_0
age, treatment, race, cd4	0.286	-39.365	0.497	0.250
age, treatment, cd4	0.487	-36.665	0.477	0.500
age, treatment, cd4	0.486	-35.366	0.499	1.000
age, treatment, cd4	0.486	-34.956	0.518	2.000

Lastly, we compute probabilities for several values of (δ_0, λ_0) and $\psi_0 = \tau_0 = \sqrt{5}$:

Prior comparison

Now, we set $\beta_0^{(m)} \sim \text{Cauchy}(0, 10)$ and $\beta_j^{(m)} \sim \text{Cauchy}(0, 2.5)$ for $j = 1, \dots, p^{(m)} - 1$ with $a_0 \sim \text{Beta}(1, 1)$.

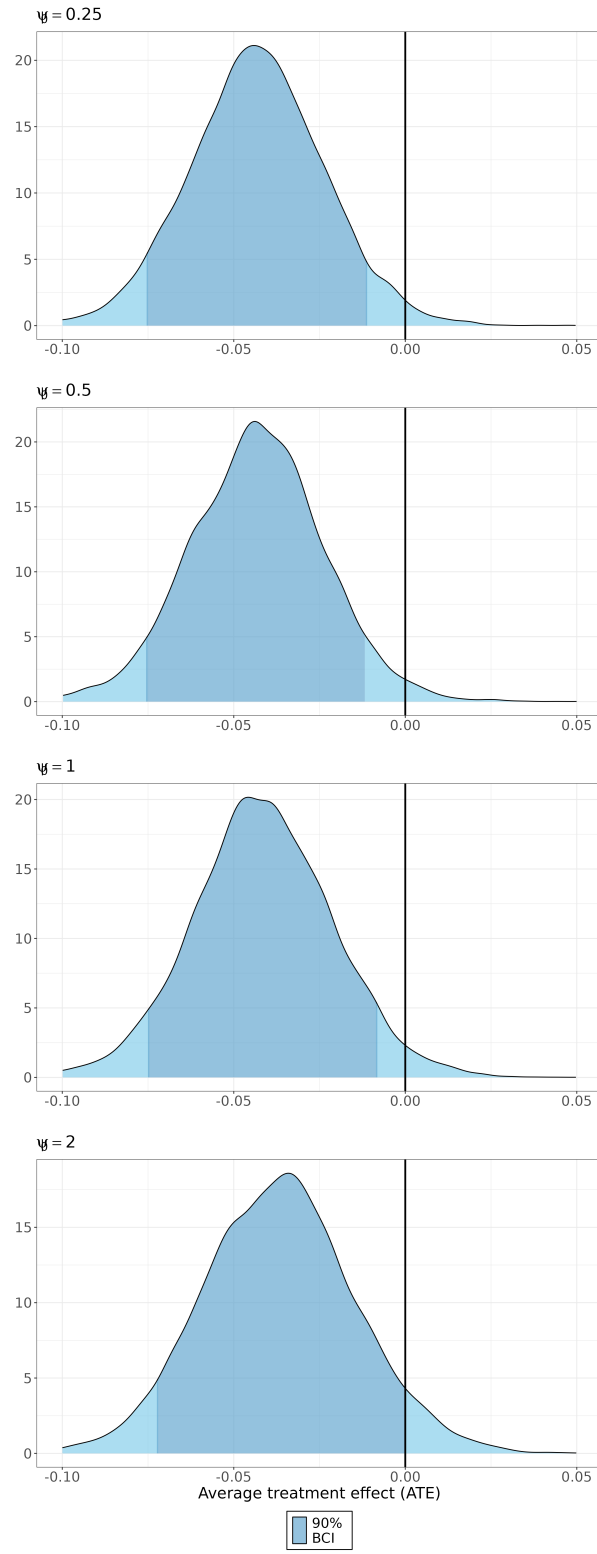


Figure 1: ATE distribution with normal prior for several values of ψ_0 .

m	$p(m)$	$\log(Z_m(D))$	$p(m D)$	(δ_0, λ_0)
age, treatment, cd4	0.487	-35.856	0.491	(1, 1)
age, treatment, cd4	0.486	-35.806	0.490	(2, 2)
age, treatment, cd4	0.487	-37.556	0.442	(1, 10)
age, treatment, cd4	0.487	-35.429	0.497	(10, 1)

m	$p(m)$	$\log(Z_m(D))$	$p(m D)$
age, treatment, cd4	0.196	-34.885	0.253
age, treatment, race, cd4	0.276	-35.509	0.191
treatment, cd4	0.085	-34.776	0.123
treatment, race, cd4	0.140	-35.357	0.113
age, race, cd4	0.131	-35.414	0.100
age, cd4	0.066	-34.746	0.099
race, cd4	0.074	-35.226	0.068
cd4	0.031	-34.577	0.054
age, treatment	0.000	-41.405	0.000
age, treatment, race	0.000	-42.075	0.000
age	0.000	-41.591	0.000
treatment	0.000	-41.758	0.000
age, race	0.000	-42.320	0.000
treatment, race	0.000	-42.392	0.000
race	0.000	-42.558	0.000

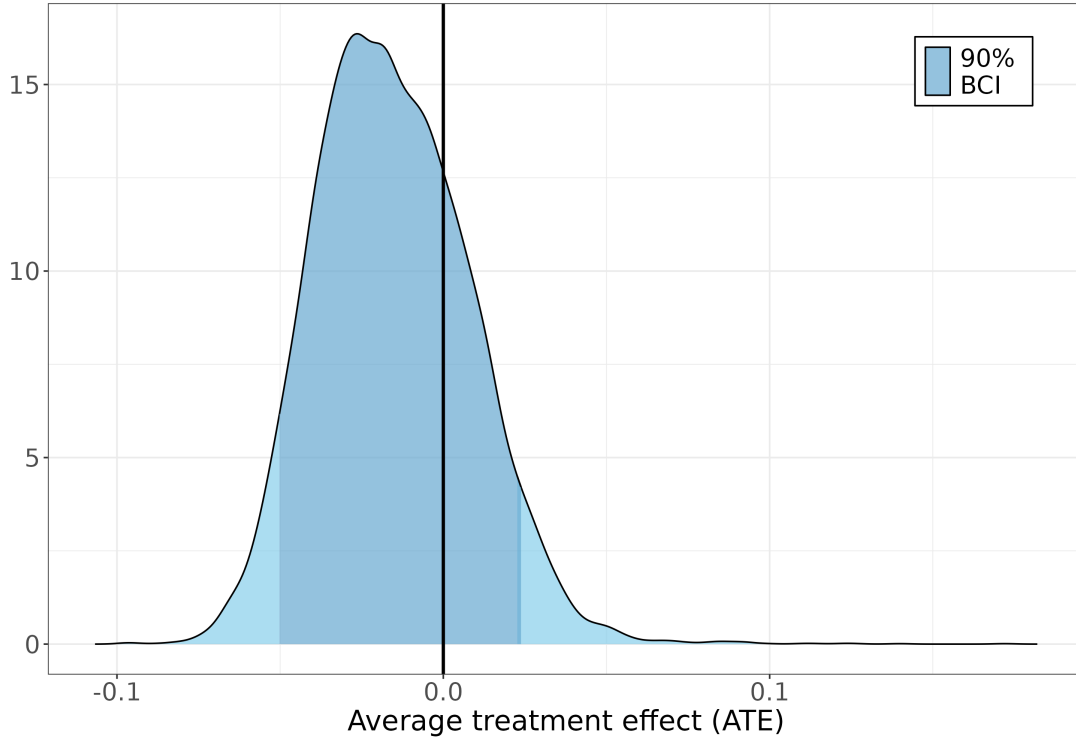


Figure 2: ATE distribution with Cauchy prior.